

UNIT-I

1. Write a program to find the largest element among three Numbers.

AIM:

To write a Python program that takes three numbers as input and finds the largest among them.

ALGORITHM:

1. Start.
2. Read three numbers a, b, and c from the user using input().
3. Compare the numbers:
 - If a is greater than or equal to both b and c, then a is the largest.
 - Else if b is greater than or equal to both a and c, then b is the largest.
 - Else, c is the largest.
4. Print the largest number.
5. End.

PROGRAM:

```
# Step 1: Take input from the user
# Convert each input to float to handle decimal numbers
a = float(input("Enter the first number: "))
b = float(input("Enter the second number: "))
c = float(input("Enter the third number: "))

# Step 2: Compare the three numbers to find the largest
if a >= b and a >= c: # If 'a' is greater than or equal to both 'b' and 'c'
    largest = a
elif b >= a and b >= c: # If 'b' is greater than or equal to both 'a' and 'c'
    largest = b
else: # Otherwise, 'c' is the largest
    largest = c

# Step 3: Display the result
print(f"The largest number among {a}, {b}, and {c} is: {largest}")
```

OUTPUT:

```
Enter the first number: 5
Enter the second number: 8
Enter the third number: 20
The largest number among 5.0, 8.0, and 20.0 is: 20.0
```

2. Write a Program to display all prime numbers within an interval.

AIM:

To write a Python program that accepts a start and end value from the user and displays all prime numbers within that range

ALGORITHM:

1. Start.
2. Read the start and end values of the interval from the user.
3. Loop from start to end (inclusive).
4. For each number in that range:
 - If the number is greater than 1:
 - Check if it has any divisors other than 1 and itself.
 - If no divisors are found, it is a prime number.
5. Print each prime number found.
6. End.

PROGRAM:

```
# Step 1: Take input for the range from the user
start = int(input("Enter the starting value of the interval: "))
end = int(input("Enter the ending value of the interval: "))

print(f"Prime numbers between {start} and {end} are:")

# Step 2: Loop through the range
for num in range(start, end + 1):
    if num > 1:
        # Step 3: Check if 'num' is prime
        for i in range(2, int(num ** 0.5) + 1):
            if num % i == 0:
                break # Not a prime number
        else:
            print(num, end=' ')
```

OUTPUT:

```
Enter the starting value of the interval: 20
Enter the ending value of the interval: 40
Prime numbers between 20 and 40 are:
23 29 31 37
```

3. Write a program to swap two numbers without using a temporary variable

AIM:

To write a Python program that swaps the values of two variables without using a third (temporary) variable.

ALGORITHM:

1. Start.
2. Read two numbers a and b from the user.
3. Swap the numbers using arithmetic operations:
 - $a = a + b$
 - $b = a - b$
 - $a = a - b$
4. Print the values of a and b after swapping.
5. End.

PROGRAM:

```
# Step 1: Take input from the user
a = float(input("Enter the first number (a): "))
b = float(input("Enter the second number (b): "))

print(f"\nBefore swapping: a = {a}, b = {b}")

# Step 2: Swap using arithmetic operations (no third variable)
a = a + b
b = a - b
a = a - b

# Step 3: Display the result
print(f"After swapping: a = {a}, b = {b}")
```

OUTPUT:

```
Enter the first number (a): 5
Enter the second number (b): 10

Before swapping: a = 5.0, b = 10.0
After swapping: a = 10.0, b = 5.0
```

4. Demonstrate the following Operators in Python with suitable examples.

- i) Arithmetic Operators
- ii) Relational Operators
- iii) Assignment Operators
- iv) Logical Operators
- v) Bit wise Operators
- vi) Ternary Operator
- vii) Membership Operators
- viii) Identity Operators

AIM:

To write Python programs that demonstrate the functionality of different types of operators: Arithmetic Operators, Relational Operators, Assignment Operators, Logical Operators, Bitwise Operators, Ternary Operators, Membership Operators, Identity Operators

PROGRAMS:

i) Arithmetic Operators

```
# Taking input from user
a = int(input("Enter first number (a): "))
b = int(input("Enter second number (b): "))

# Performing and displaying arithmetic operations
print(f'Addition: {a + b}')
print(f'Subtraction: {a - b}')
print(f'Multiplication: {a * b}')
print(f'Division: {a / b if b != 0 else 'Undefined (division by zero)'}')
print(f'Floor Division: {a // b if b != 0 else 'Undefined'}')
print(f'Modulus: {a % b if b != 0 else 'Undefined'}')
print(f'Exponentiation: {a ** b}')
```

OUTPUT:

```
Enter first number (a): 6
Enter second number (b): 9
Addition: 15
Subtraction: -3
Multiplication: 54
Division: 0.6666666666666666
Floor Division: 0
Modulus: 6
Exponentiation: 10077696
```

ii) Relational Operators

```
# Taking input from user
a = int(input("Enter first number (a): "))
b = int(input("Enter second number (b): "))

# Performing relational comparisons
print(f'a == b: {a == b}') # Equality
print(f'a != b: {a != b}') # Not equal
print(f'a > b: {a > b}') # Greater than
print(f'a < b: {a < b}') # Less than
print(f'a >= b: {a >= b}') # Greater than or equal
print(f'a <= b: {a <= b}') # Less than or equal
```

OUTPUT:

```
Enter first number (a): 10
Enter second number (b): 23
a == b: False
a != b: True
a > b: False
a < b: True
a >= b: False
a <= b: True
```

iii) Assignment Operators

```
# Input from user
a = int(input("Enter a number: "))
x = a # Basic assignment

# Performing compound assignments
print(f'x = {x}')
x += 5 # Add and assign
print(f'x += 5 = {x}')
x -= 2 # Subtract and assign
print(f'x -= 2 = {x}')
x *= 3 # Multiply and assign
print(f'x *= 3 = {x}')
x //= 4 # Integer divide and assign
print(f'x //= 4 = {x}')
x %= 3 # Modulus and assign
print(f'x %= 3 = {x}')
```

OUTPUT:

```
Enter a number: 8
x = 8
x += 5 = 13
x -= 2 = 11
x *= 3 = 33
x //= 4 = 8
x %= 3 = 2
```

iv) Logical Operators

```
# User inputs boolean values as 0 or 1
a = bool(int(input("Enter first boolean (0 or 1): ")))
b = bool(int(input("Enter second boolean (0 or 1): ")))
```

```
# Performing logical operations
print(f'a and b → {a and b}') # Logical AND
print(f'a or b → {a or b}')   # Logical OR
print(f'not a → {not a}')     # Logical NOT
```

OUTPUT:

```
Enter first boolean (0 or 1): 0
Enter second boolean (0 or 1): 1
a and b → False
a or b → True
not a → True
```

v) Bitwise Operators

```
# Taking integer input from user
a = int(input("Enter first integer: "))
b = int(input("Enter second integer: "))

# Performing bitwise operations
print(f'a & b = {a & b}') # Bitwise AND
print(f'a | b = {a | b}') # Bitwise OR
print(f'a ^ b = {a ^ b}') # Bitwise XOR
print(f'~a = {~a}')       # Bitwise NOT
print(f'a << 1 = {a << 1}') # Left shift
print(f'a >> 1 = {a >> 1}') # Right shift
```

OUTPUT:

```
Enter first integer: 45
Enter second integer: 23
a & b = 5
a | b = 63
a ^ b = 58
~a = -46
a << 1 = 90
a >> 1 = 22
```

vi) Ternary Operator

```
# Taking input from user
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

# Using ternary to find maximum
max_val = a if a > b else b
print(f"The greater number is: {max_val}")
```

OUTPUT:

```
Enter first number: 5
Enter second number: 10
The greater number is: 10
```

vii) Membership Operators

```
# Predefined list
sample_list = [10, 20, 30, 40, 50]

# Input to check for membership
item = int(input("Enter an item to check in the list: "))

# Checking membership
print(f"{item} in list = {item in sample_list}")
print(f"{item} not in list = {item not in sample_list}")
```

OUTPUT:

```
Enter an item to check in the list: 50
50 in list = True
50 not in list = False
```

viii) Identity Operators

```
# Defining three lists
x = [1, 2, 3]
y = x          # Same reference
z = [1, 2, 3]  # Same values, different object

# Identity and equality checks
print(f'x is y = {x is y}')  # True: same object
print(f'x is z = {x is z}')  # False: different object
print(f'x == z = {x == z}')  # True: same content
```

OUTPUT:

```
x is y = True
x is z = False
x == z = True
```

5. Write a program to add and multiply complex numbers

AIM:

To write a Python program that performs addition and multiplication of two complex numbers entered by the user.

ALGORITHM:

1. Start.
2. Read two complex numbers c1 and c2 from the user using complex() function.
3. Perform:
 - Addition: $c1 + c2$
 - Multiplication: $c1 * c2$
4. Display the results.
5. End.

PROGRAM:

```
# Step 1: Input complex numbers from the user
# The user should enter numbers in the form a+bj (e.g., 3+4j)
c1 = complex(input("Enter the first complex number (e.g., 3+4j): "))
c2 = complex(input("Enter the second complex number (e.g., 1+2j): "))

# Step 2: Perform addition and multiplication
sum_result = c1 + c2
product_result = c1 * c2

# Step 3: Display the results
print(f"\nAddition of {c1} and {c2} is: {sum_result}")
print(f"Multiplication of {c1} and {c2} is: {product_result}")
```

OUTPUT:

```
Enter the first complex number (e.g., 3+4j): 3+7j
Enter the second complex number (e.g., 1+2j): 3+2j

Addition of (3+7j) and (3+2j) is: (6+9j)
Multiplication of (3+7j) and (3+2j) is: (-5+27j)
```

6. Write a program to print multiplication table of a given number.

AIM:

To write a Python program that takes a number and a limit from the user, and prints the multiplication table of the number up to the specified limit.

ALGORITHM:

1. Start
2. Input a number n from the user.
3. Input the range limit (i.e., up to how many multiples you want).
4. Use a loop from 1 to limit:
 - a. Multiply n by the loop counter i.
 - b. Print the result in the format n x i = result.
5. End

PROGRAM:

```
# Step 1: Take input from the user
n = int(input("Enter the number: "))
limit = int(input("Enter the range up to which you want the multiplication table: "))

# Step 2: Print the multiplication table
print(f"\nMultiplication Table of {n} up to {limit}:\n")

for i in range(1, limit + 1):
    print(f"{n} x {i} = {n * i}")
```

OUTPUT:

```
Enter the number: 10
Enter the range up to which you want the multiplication table: 10

Multiplication Table of 10 up to 10:

10 x 1 = 10
10 x 2 = 20
10 x 3 = 30
10 x 4 = 40
10 x 5 = 50
10 x 6 = 60
10 x 7 = 70
10 x 8 = 80
10 x 9 = 90
10 x 10 = 100
```